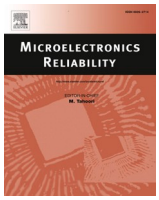
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Ageing metallized polypropylene film capacitors laws confronted with the phenomenon of corrosion

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A B S T R A C T

Metallized polypropylene film capacitors are known to be one of the most common causes of failure in electronic systems. Predicting their lifetime to anticipate failures is a key issue in the assessment of these systems' reliability. In this paper, accelerated ageing tests applying voltage, temperature and humidity were conducted on 42 capacitors. The aim is to evaluate and sharpen the existing laws in literature thanks to the support of these data. Therefore, based on an analysis of failures and the understanding of this phenomenon, a new law is introduced modelling the capacitance degradation. The model has been assessed against the data, as compared to present laws in literature and other functions, describing the evolution of experimental curves.

1. Introduction

Metallized film capacitors are critical components, which are widely used in power electronics systems. Considering all types of capacitors, they are responsible for 30 % of the systems failures [1]. Predicting their lifetime is a key point in their reliability enhancement. This will contribute to the safety improvement of said systems, thereby increasing their proper functioning as well as their maintenance optimization.

A first series of methodologies are focused on lifetime forecasts based on stochastic models [2–4] modelling the capacitance loss. However, most of the time, this loss of capacitance and its occurrences are considered consistent over time, which is not always accurate, depending on the mechanism's failure. Other studies have proposed empirical ageing laws, based on the degradation of key indicators such as capacitance and ESR (Equivalent Series Resistance) [5] or on the quantification of the influence of stress factors such as temperature, humidity and voltage [6,7]. The latter may not be resistant to changes in ageing conditions or manufacturing processes. Moreover, the expected lifetime must be derived from accelerated ageing tests under known environmental conditions. Finally, recent studies proposed some laws based on the failures physics for specific phenomenon [8], which must be adapted according to the data.

As a rule, the capacitor's end of life is limited the moment where the capacitance decreases from 5 % to 20 % of its initial value [9]. In fact, this limit depends on the principal function of the capacitors in the system. Therefore, this limit can be increased.

The objective of this study is to evaluate the literature laws focusing on the accelerated ageing data and sharpen them if necessary. Firstly, the results of accelerated ageing tests beyond the usual limits will be analysed. Secondly, models based on mechanism failure phenomenon will be presented with the aim of describing the capacitance reduction through ageing. Finally, the existing functions of literature will be compared.

2. Failure mechanisms

Available literature has reported two main gradually deteriorating mechanism failures through the recording of metallized capacitors: the corrosion and the self-healing. The corrosion is due to the oxidation of the capacitor electrodes mainly composed of aluminium and zinc according to the chemical reactions below (Eqs. (1) to (4)) [10,11]:

$$2Al + 3H_2O \rightarrow Al_2O_3 + 3H_2 \quad (1)$$

$$2Al + 6H_2O \rightarrow 2Al\,(OH)_3 + 3H_2 \quad (2)$$

$$2Zn + O_2 \rightarrow 2ZnO \quad (3)$$

$$2Zn + 2H_2O + O_2 \rightarrow 2Zn(OH)_2 \quad (4)$$

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The AC voltage reinforces the phenomenon by enhancing the electrolysis, in particular the water electrolysis (Eqs. (5) and (6)).

$$2H_2O \rightarrow O_2 + 4H^+ + 4e^- \quad (5)$$

$$4H_2O + 4e^- \rightarrow 2H_2 + 4OH^- \quad (6)$$

The presence of water nearby the metallization creates electrochemical cells where reactions occur. The AC voltage applied between the two electrodes of the capacitor will polarize the dielectric along with the moisture/metallization interface [11]. This is visible on the metallization as spots. By decreasing potential standard, the redox couples involved in the reactions are O_2/H_2O, O_2/HO^-, H^+/H_2, Zn^{2+}/Zn, H_2O/H_2, Al^{3+}/Al. The AC voltage accelerates the formation of reactants like hydroxide ions and dioxygen through the decomposition of water molecules. Then these reactants will either directly react with the metals, the aluminium and the zinc, to oxidize them directly or go through a redox reactions to ionize them into Zn^{2+} and Al^{3+} to make them react with the hydroxide ions HO^- and finally produce the associated oxides [8].

The reduction and the oxidation of water molecules or their ionized form is easily reversible thanks to electrolysis unlike the oxidation of aluminium or zinc [11]. The oxidized metal layer formed slows down the reaction rate. The increase of its thickness reduces the access to the metallization for hydroxide ions. The process is enhanced by the applied voltage, the humidity, and the temperature. The voltage applies an external driving force helping the ions to penetrate the oxidized metal layer and, thus, to extend the corrosion spots [12]. The temperature increases the speed of each chemical reaction.

Self-healing is a phenomenon that occurs in dielectric materials, when a partial discharge occurs between two electrodes because of an impurity located in the dielectric. The discharge locally heats the capacitor while vaporizing the impurities and the metallization near the defect [12] as illustrated Fig. 1. Therefore, the dielectric strength is restored. This phenomenon is possible, thanks to the thickness of the metallization (~nm) that is 1000 times thinner than the thickness of the dielectric (~μm).

The discharge breaks the molecular bonds of the dielectric. It leads to the formation of stable gases and leaves a symmetric hole in the dielectric on each side of the electrode, as illustrated in Fig. 2. For the polypropylene, the gases formed are CO, H_2, C_2H_2, CH_4.

The discharge melts the dielectric locally where the impact took place as shown on the film in Fig. 1. This is the main difference that distinguishes a de-metallized circular spot coming from corrosion or from self-healing. This phenomenon is enhanced also by the applied voltage, the humidity, and the temperature. It results in decreasing the dielectric resistance [13].

3. Tests conditions

The tested capacitors are metallized polypropylene film capacitors with reference value of 2.2 μF, 330 V AC rated voltage (Vr), used to filter electromagnetic interference. In total 42 capacitors were aged from 3 different manufacturers. The capacitors from each manufacturer were divided into groups of 7 for two different tests over a 3-month period under fixed conditions of temperature, voltage, and humidity. The conditions, detailed in Table 1, were chosen according to the highest

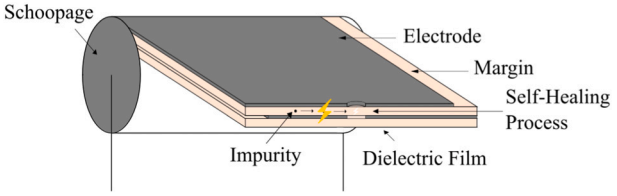


Fig. 1. Self-healing process on metallized film capacitor.

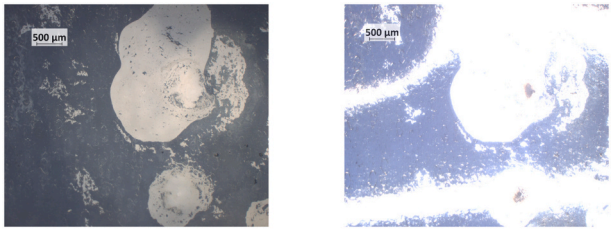


Fig. 2. Self-healing spots visible with optical microscope.

Table 1
Experimental conditions.

	Temperature	Relative humidity	Voltage
Test 1	85 °C	85 %	1.Vr (330 V AC 50 Hz)
Test 2	85 °C	85 %	1,1.Vr (363 V AC 50 Hz)

expected degradation of capacitance without causing any other unusual failure mechanisms. These conditions were defined based on the literature data tests [4,8]. As the effect of the humidity on capacitance has already been studied in previous papers, in our study we chose to vary the voltage to observe other forms of capacitance degradation.

The AC voltage was applied through a three-phase autotransformer. Seven capacitors from the same manufacturer were placed in parallel between 2 phases of the supply. An inverter was used upstream to guarantee the stability of the applied voltage and its frequency of 50 Hz. The voltage and the current were sensed for each capacitor at 1 kHz with an accuracy of <2.5 % and 2 % respectively. A view of the test bench is shown in Fig. 3.

The RMS current I_{rms} and voltage U_{rms} were calculated from the root mean square of the corresponding signals over a period of one second. By considering the pulsation 'ω' constant, thanks to the fixed frequency at 50 Hz, the capacitance has been calculated by neglecting the ESR as below:

$$C = \frac{Irms}{Urms.\omega} \quad (7)$$

The dielectric resistance was measured with a megohmmeter with an accuracy of 20 %, one to three times per week. The capacitors were individually disconnected from the applied voltage to perform the measurement, while remaining in the climate chamber with the test conditions of 85 °C and 85 % relative humidity. The dielectric resistance was measured after the application of 100 V DC for the duration of 1 min. These values have been chosen based on the IEC-60384-1 standard.

Throughout all the testing and measurement periods, the capacitors remained under the test temperature and humidity conditions.

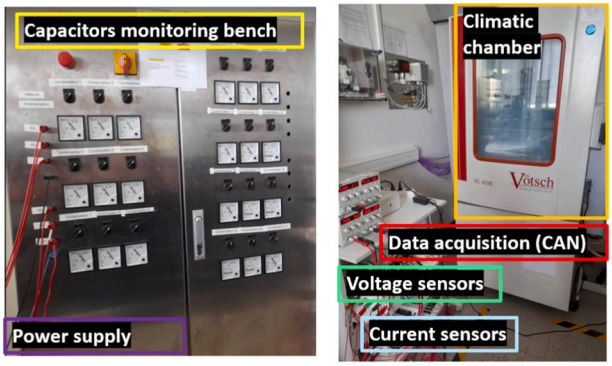


Fig. 3. View of the test bench.

4. Results

The curves representing the capacitance degradation of the devices from the different manufacturers are presented in Fig. 4. They have a similar shape to those presented in another study, including similar tests carried out at the same temperature of 85 °C and nominal voltage, but with lower relative humidity of 60 % and 30 % [8]. By decreasing the humidity level, the curves stretch according to a scale factor. In this paper, the increase of the voltage between the two different tests accelerates the degradation without changing the shape of the curves. It does not impact the dynamics of the degradation. Therefore, the impact of voltage on the ageing curve can also be modelled by a scale factor.

ΔC represents capacitance evolution over time divided by the initial capacitance ($C_{initial}$).

$$\Delta C(t) = \frac{C(t)}{C_{initial}} \tag{8}$$

For the first ten to hundred hours of ageing depending on the manufacturer, the capacitance stays relatively steady. During this period, the capacitors have almost not been affected by the stress factors.

Then, the degradation rate increased. Some studies propose an exponential function to model this phase [5] under similar environmental conditions of ageing. It is illustrated by Eq. (9) with A_9, B_9 and α_9 as positive constants.

$$C(t) = A_9 - B_9.exp\ (\alpha_9.t) \tag{9}$$

Other studies propose a variant for this phase, based on the corrosion process and leading to the Eq. (10) [8]. A_{10}, B_{10} and m are positive constants.

$$C(t) = A_{10} - B_{10}.t^m \tag{10}$$

However, in the next phase, the degradation slows down. The capacitance degradation converges towards a residual value. This slowdown is consistent with an inverse exponential function described in some studies [4], usually used to describe capacitance losses from self-healing with A_{11}, B_{11} and α_{11} as positive constants.

$$C(t) = A_{11} + B_{11}.exp\ (-\alpha_{11}.t) \tag{11}$$

However, neither of these models describes the complete behaviour of the capacitance degradation by ageing.

The variation of the dielectric resistance is presented in Fig. 5 over the ageing period. The curves show the impact of the humidity, temperature, and voltage on the dielectric strength making the self-healing mechanisms more likely to happen. Moreover, the curves are correlated to the capacitance degradation curves. They also show a strong decrease at about the same ageing time. This could be because of humidity diffusion in the dielectric. It should be noted that the first measurement was performed after the first hundred hours of ageing. Therefore, the curves in Fig. 5 don't contain a stable phase.

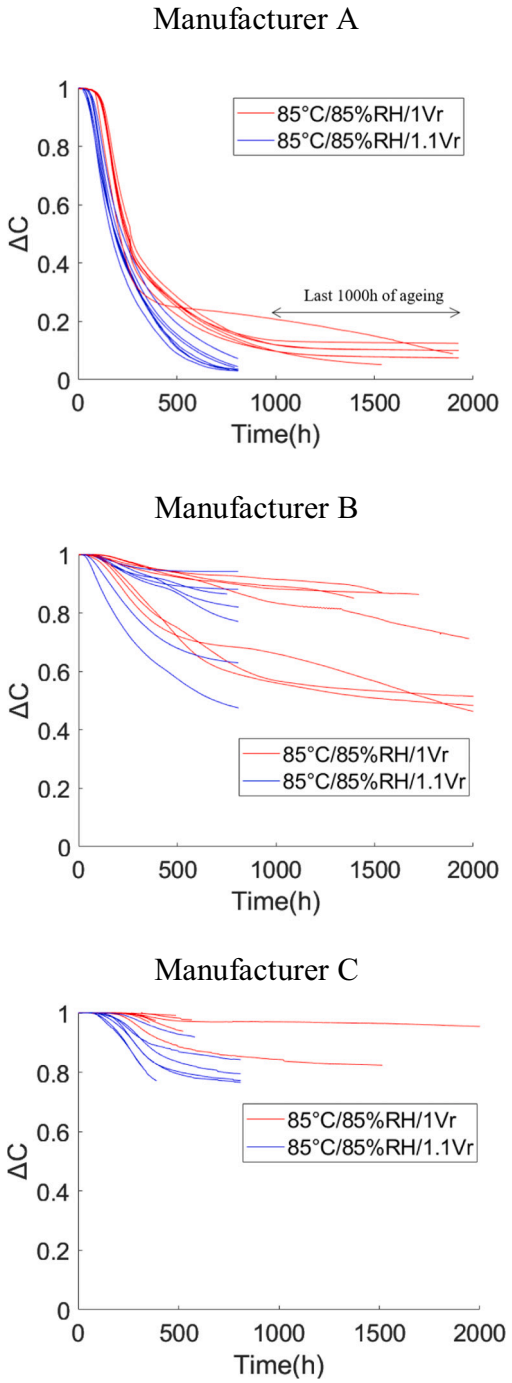


Fig. 4. Capacitance degradation on capacitors from different manufacturers.

5. Post-mortem failure analysis

The state of the metallized films after the first ageing test is shown in Fig. 6. Two metallized films initially placed one on top of the other were separated and placed head-to-tail for each manufacturer. The films come from the middle part of the capacitor winding. Their appearance is representative of the general state of the entire capacitor, starting from the outer to the inner windings.

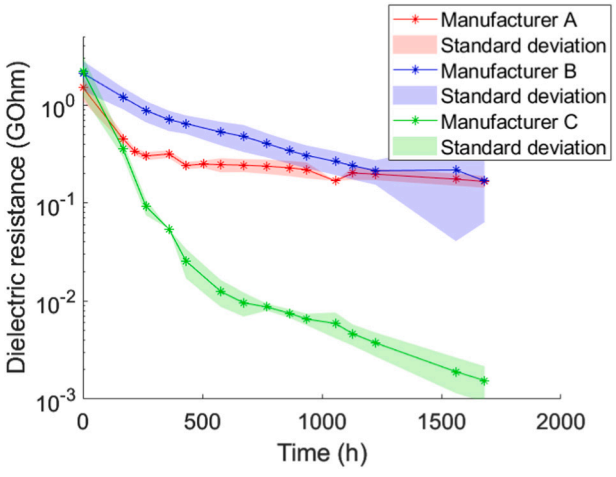


Fig. 5. Dielectric resistance degradation from different manufacturers for the first test at 85 °C/85 %RH/Vr.

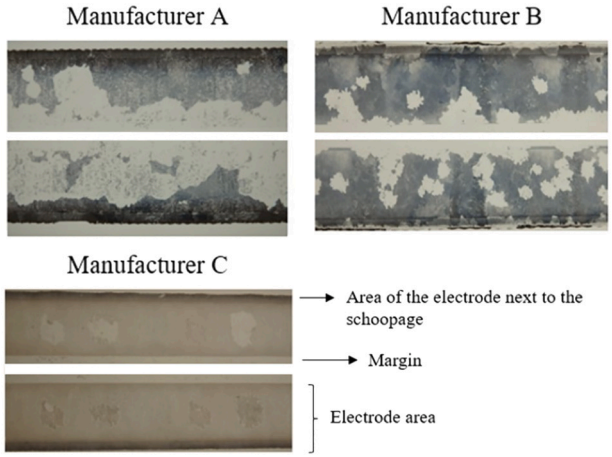


Fig. 6. State of the films of a capacitor from the three manufacturers after the first ageing test period (cf Table 1).

As observed, the reduction in capacitance is due to the reduction of the electrode area. The metallization has been heavily attacked on the edge close to the electrode margin, for the manufacturers A and B. The metallized film of the manufacturer C was attacked less, and this is consistent with their relatively small reduction in capacitance up to 25 % for 2000 h. In comparison, all capacitors of the manufacturer A reach up to 80 % of capacitance degradation. The devices from manufacturer B show more heterogeneous degradation. However, half of the devices' capacitance degradation exceed 25 %, some of them reaches 50 %.

The demetallized areas appear as non-symmetrical spots on each side of the electrodes. On the film, there are no hole marks to suggest that an electrical discharge occurred. The images obtained by microscope show no sign of melted spots in the dielectric. Conversely, there are variations in the colour intensity of the metallization, specifically close to the demetallized spots. This means that the demetallization process of the area has been gradual. All these elements clearly show that corrosion was the main failure mechanism. When the metallization area is intact, its colour appears black. In case that oxidation occurs through its thickness, the colour changes to a grey tint and finally disappears, so turns to white in Fig. 6, once the metallization is completely oxidized. If the metallization area is vaporized through a self-healing, it disappears instantaneously completely without changing before to a grey tint. There are several sizes of corrosion spots. In Fig. 6, the smaller ones have spread and overlapped to form the larger ones.

Symmetrical spots are visible on the metallized film from manufacturer C. Some areas are completely demetallized and appear white, while other areas appear darker, still covered by metallization. There is not any sign of discharges in the centre of these spots. The most plausible explanation is the transfer of metallization from one film to another placed underneath during the unwinding of the capacitor. For capacitors from manufacturer C, the oxidation did not degrade completely the metallization. One possible explanation could be a bigger thickness or a more robust metallization composition to corrosion.

It should be noted that the edge of the electrode close to the schoopage connection for each manufacturer appears in black, meaning the impact of oxidation for this area has been minimal. This is consistent with the electrode profile design introduced by manufacturers, which often have thick edges in this area to reduce the electrical losses, increase the current rating [14] and grip better the schoopage.

6. Modelling the corrosion impact on capacitance

As demonstrated above, the main ageing mechanism in our study is the corrosion. In case that self-healing phenomena have occurred, it has a minor impact. It can therefore be disregarded. As it is mentioned in Eq. (12), the capacitance is a function of the permittivity of the dielectric $\varepsilon_0\varepsilon_r$, the surface of the electrodes A where the electrical charges are stored, also named the active area, and d, the distance between the electrodes.

$$C(t) \approx \frac{\varepsilon_0 \varepsilon_r A}{d}(t) \tag{12}$$

The permittivity of the dielectric will be considered consistent in the model presented in this paper, since the test conditions are stable. As the corrosion will affect the thickness by reducing the metallization itself, the distance between the electrodes d will slightly increase. However, this can be neglected as in metallized film capacitors, the thickness of the dielectric (~ μm) is about 1000 times greater than the thickness of the metallization (~ nm). Therefore, the distance between the electrodes d is the thickness of the dielectric. Therefore, this distance will be assumed to remain consistent. Consequently, it is assumed that the reduction in capacitance comes solely from the reduction of electrodes' surface caused by the corrosion.

According to the Wagner's theory of metal oxidation [15], the thickness growth of the oxide is proportional to the square of the time. The speed of the oxidation process decreases with the oxide layer growth. As oxide thickness is larger, the ions travel a longer distance before reaching the metal and reacting with it. As the metallization thickness has much smaller dimension in length (~ nm) compared to its width (~ 2 cm) and length (~ 50 m), it is assumed that corrosion kinetics will depend primarily on the oxide growth through it. It is also assumed that moisture diffusion takes place in the space between the different metallized film layers. These metallized film layers consist of a dielectric layer and a metal layer.

The kinetics of oxide growth, since it is caused by corrosion, depends on the associated stress factors, i.e., temperature, voltage, and humidity. It also depends on the oxide formed itself i.e., the metallic composition of the electrodes. The duration of oxide growth through the metallization thickness is associated to a frequency λ, which is the destruction frequency of the associated metallization area.

The total active area of the electrode A(t) can be considered as the sum of several area units associated with their own destruction frequency λ_i by corrosion. If λ is constant and the same for each unit area ($\lambda_i = \lambda$), the total surface evolution will follow the exponential law below:

$$\frac{dA(t)}{dt} = -\lambda.A(t) \tag{13}$$

$$\Rightarrow A(t) = A_{initial}.exp(-\lambda.t) \tag{14}$$

$A_{initial}$ is the active area of the electrode before ageing.

For the model shown below, it is assumed that the thickness and metal composition of the electrode are identical at every point on its surface for each capacitor of each manufacturer. As the applied voltage will remain constant throughout the test, across all the capacitor layers, λ_i will be considered consistent during ageing. Similarly, the impact of temperature on the λ_i can be interpreted by considering the time it takes to reach the test temperature homogeneously inside the capacitor, trivial compared to the ageing time (of the order of one hour for test durations of 1000 h). On the other hand, the impact of moisture diffusion must be considered. In view of the above figures, the λ_i depends only on the moisture diffusion over time and space. A final assumption to simplify the model could be done by considering that λ_i are independent of their position on the electrode (close or not to the electrode's edge) and equal to their average value $\lambda(t)$ which will be defined in the next paragraph.

Before being put into service, the capacitor could contain a certain degree of moisture, which may have been caused either by impurities that slipped between the films during the manufacturing process, or by the ambient moisture that has been diffused out through the layers during storage. This low humidity level leads to a fixed initial failure rate of λ_0. After complete diffusion of the moisture through the capac-

itor's layers, this rate increases to a fixed value of λ_1. The surface of the electrodes is a finite surface where each unit area has either a failure rate of λ_0 or λ_1 depending on whether moisture has diffused to it or not. $A_{\lambda_0}(t)$ is the area of the electrode containing all units which have a λ_0 failure rate. Consequently, with $\alpha(t)$ the percentage of area infected by λ_1, the average value $\lambda(t)$ of per unit area is:

$$\lambda(t) = \lambda_0.(1 - \alpha(t)) + \lambda_1.\alpha(t) \tag{15}$$

$$\text{with } \alpha(t) = \frac{A_{initial} - A_{\lambda_0}(t)}{A_{initial}} \tag{16}$$

Considering that moisture propagates from a specific area unit and contaminates surrounding surfaces gradually from λ_0 to λ_1, the moisture diffusion is equivalent to the spread out of an epidemic. Therefore, by analogy, the famous SI (susceptible-infected) model can be used. We obtain the following relationship where β is the moisture diffusion rate:

$$\frac{dA_{\lambda_0}(t)}{dt} = -\beta.A_{\lambda_0}(t)\left(A_{initial} - A_{\lambda_0}(t)\right) \tag{17}$$

$$\Rightarrow \frac{d(1-\alpha(t))}{dt} = -\beta.\alpha(t)\,(1-\alpha(t)) \tag{18}$$

Combining this result with Eq. (15) give the sigmoid function:

$$\lambda(t) = \lambda_0 + \frac{\lambda_1 - \lambda_0}{1 + D.exp(-\beta.t)} \tag{19}$$

where $D = \frac{1-\alpha(0)}{\alpha(0)}$.

Theoretically D is infinite. Mathematically, to solve Eq. (18) above and remain within the definition domain of functions, $\alpha(0)$ should be different from 0. However, it could be the case when any area has a failure rate of λ_1. To solve this problem, $\alpha(0)$ will be approximated to a small fixed value close to 0. For the simulation $\alpha(0)$ will be fixed here at 0.01 %.

Unlike the self-healing phenomena, the areas destroyed by corrosion are not symmetrical on each side of the capacitor's electrode. The metallization's active surface A as used in the capacitance calculation (Eq. (12)), is the intersection of the superposition of the upper and lower conductive surfaces of the electrodes. Consequently, the oxidized surface of the active electrode is the common oxidized surfaces of both electrodes as shown in Fig. 7. Therefore, the failure rate $\lambda_a(t)$ of this specific area unit, is the probability to lose this specific area either on the upper or on the lower electrode and it is described by the equation below:

$$\lambda_a(t) = 2\,\lambda(t) - \lambda(t)^2 \tag{20}$$

with $\lambda(t)$, the failure rate of this specific area unit for both top and bottom electrodes, the probability $\lambda(t)$ of losing a specific area on one of the electrodes being independent of the loss of this same specific area on the other electrode.

As a result, the corrosion process affects the capacitance according to the Eqs. (12), (14), (19) and (20) as:

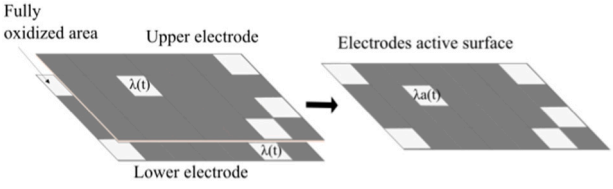


Fig. 7. Impact of corrosion damage on the active area.

$$C(t) = C_{initial}.exp\left(-\left(2\left(\lambda_0 + \frac{\lambda_1 - \lambda_0}{1 + D.exp(-\beta.t)}\right) - \left(\lambda_0 + \frac{\lambda_1 - \lambda_0}{1 + D.exp(-\beta.t)}\right)^2\right).t\right) \tag{21}$$

7. Interpolation

The model introduced in this paper (Eq. (21)) will be assessed against the data in this study and compared with the functions proposed in the literature (Eqs. (9), (10) and (11)) and other relevant functions described below. As established above, the functions proposed in the literature do not consider both the acceleration and deceleration phases of capacitance degradation, but only one of them. Consequently, a combination of these functions will be tested (Eq. (22)) as well as a generic function of the data form which is the sigmoid function (Eq. (23)).

$$C(t) = A_{22} - B_{22}.exp\,(\gamma_{22}.t) + E_{22}.exp\,(-\alpha_{22}.t) \tag{22}$$

$$C(t) = A_{23} - \frac{1}{B_{23} + E_{23}.exp\,(-\gamma_{23}.t)} \tag{23}$$

To compare the latter with bibliographic functions for which three coefficients must be determined, one coefficient should be fixed. Per Eq. (19) the delay coefficient E_{23} expressing the shift of the sigmoid function will be fixed with the same value as D.

$$C(t) = A_{24} - \frac{1}{B_{24} + D.exp\,(-\gamma_{24}.t)} \tag{24}$$

For Eqs. (22) to (24), A_{22}, B_{22}, E_{22}, γ_{22}, α_{22}, A_{23}, B_{23}, E_{23}, γ_{23}, A_{24}, B_{24} and γ_{24} are positive constants.

In the assumption developed above, it is worth noting that $\frac{\varepsilon_0\varepsilon_r}{d}$ is consistent throughout ageing, considering the active area of the electrode A to be the sum of small area units. Therefore, the capacitance of the capacitor is the sum of small unit capacitances corresponding to small unit capacitors connected in parallel. Consequently, it is possible to model the capacitance's degradation through ageing, by considering theses small unit capacitances failing one after the other following a probability law. Therefore, the Weibull's law is interesting and should be tested [3]. It is a generic function that models approximately a wide range of failure behaviours. It leads to the equation below, with α_{25} and n positive constants.

$$C(t) = C_{initial}.exp\left(-\left(\frac{t}{\alpha_{25}}\right)^n\right) \tag{25}$$

The interpolation of the real data with the different functions proposed was performed using an optimization algorithm [16], aiming at minimizing the mean square error (MSE). The results are presented in Table 2, considering the MSE value as the average MSE value for each of 21 capacitors for both tests. The fitting of the functions for one capacitor with a representative ageing curve and with a capacitance severely degraded is showed in Fig. 8.

Since the curves display a deceleration phase of the capacitance degradation over a longer period than the acceleration phase, the inverse exponential function (Eq. (11)) is among all bibliographic reviews, the one that has the best fit. The power law and the exponential law give the worst results (Eqs. (9) and (10)). As expected, the exponential (Eq. (9)) only adjusts well to low capacitance degradation. As it was extracted from capacitors that were aged for shorter period (up to 25 % of capacitance loss), the less accurate adjustment is entirely due to its empirical origin [5]. The sum of the exponentials (Eq. (22)) has a better fit, but also the largest number of coefficients. Its deviation from the inverse exponential law (Eq. (11)) is small and therefore irrelevant. The Weibull function (Eq. (25)) has a good ratio between its MSE and its low number of coefficients. Such as the exponential inverse function (Eq.

Table 2
MSE for the tested functions.

Functions		Total MSE (μF)	Total MSE[a] (μF)
$C(t) = A_9 - B_9.exp(\alpha_9.t)$	(9)	0.0481	0.0413
$C(t) = A_{10} - B_{10}.t^m$	(10)	0.0451	0.0593
$C(t) = A_{11} + B_{11}.exp(-\alpha_{11}.t)$	(11)	0.0020	0.0027
$C(t) = C_{initial}.exp\left(-\left(2\left(\lambda_0 + \frac{\lambda_1-\lambda_0}{1+D.exp(-\beta.t)}\right) - \left(\lambda_0 + \frac{\lambda_1-\lambda_0}{1+D.exp(-\beta.t)}\right)^2\right)t\right)$	(21)	0.0039	0.0025
$C(t) = A_{24} - \frac{1}{B_{24}+D.exp(-\gamma_{24}.t)}$	(24)	0.0133	0.0133
$C(t) = A_{23} - \frac{1}{B_{23}+E_{23}.exp(-\gamma_{23}.t)}$	(23)	0.0023	0.0019
$C(t) = A_{22} - B_{22}.exp(\gamma_{22}.t) + E_{22}.exp(-\alpha_{22}.t)$	(22)	0.0019	0.0026
$C(t) = C_{initial}.exp\left(-\left(\frac{t}{\alpha_{25}}\right)^n\right)$	(25)	0.0049	0.0049

[a] Last 1000 h of ageing the capacitors of the manufacturer A, first test (cf Table 1), removed.

(11)) and our model (Eq. (21)), the Weibull function (Eq. (25)) is based on the determination of the failure rate of small capacitance units composing the whole capacitance. The Weibull's function approximates the failure rate, while the exponential inverse function considers it consistent (cf Eq. (14)). Therefore, the comparable adjustment of these functions was predictable. These functions fit better than the generic sigmoid function (Eq. (24)), which a priori seems best suited to our data.

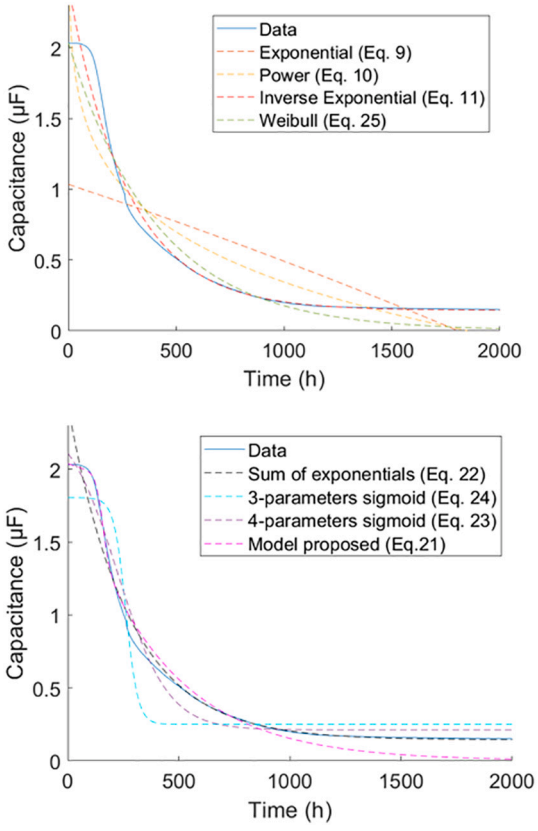


Fig. 8. Interpolation of the capacitance's degradation curve of a capacitor after the second ageing test (cf Table 1) with the different functions listed in the Table 2.

Consequently, it validates the capacitance failure modelling relevance based on the failure of small capacitance units.

The model proposed in this paper (Eq. (21)) has a higher MSE than the inverse exponential function (Eq. (11)) when it is applied to calculate the fit of the data throughout the capacitor ageing. However, according to the plotted curves representing the different models in Fig. 8, our model tends to fit better at the beginning of the ageing period while Eq. (11) fits better only at the end. It should be noticed that physically the capacitance degradation function should tend towards 0. Mathematically, Eq. (11) tends towards a constant value (A_{11}), which could be different from 0 while Eq. (21) tends towards 0 in all the cases.

As shown in Fig. 4, the capacitance degradation of some capacitors converges to a value that is not 0 during these tests. These capacitors did not reach their end of life. This stabilization of the capacitance to a value different from 0 during a long period of ageing will not be expressed by our model. This phenomenon could be due to the heterogeneities of the thickness or moisture along the electrode surface. These specific explanations will be discussed in detail in the next section. It is particularly noticeable for capacitors of manufacturer A from the first test. For these capacitors, the discrepancy is certainly due to the edges of the electrodes near the schoopage not having completely disappeared. As discussed above, it is highly probable that the metallization in this area is thicker. The limits of our model stem from these assumptions and the possibility for Eq. (11) to converge to a value other than 0 makes Eq. (11) mathematically more suitable than our model (Eq. (21)).

The purpose of the ageing laws is to fit the data from the beginning of the ageing and to be able to predict it. Adjusting to the measured points at the beginning of the degradation is more important than adjusting at the end of it. As mentioned above, Eq. (11) has a better fit only because it adjusts well to the phase of stabilization of the capacitance noted at the end of our tests. Therefore, our model, which adapts better at the beginning of the ageing phase, is more relevant. To prove it, the MSE of the functions has been recalculated by removing the last 1000 h of ageing for manufacturer A from the first test (cf. Table 1). It corresponds to the last 20 % of remaining capacitance. The results are presented in the right column of the Table 2. Unsurprisingly, in this condition, the model proposed has a better match for the same number of coefficients that must be determined. The four-parameters sigmoid function (Eq. (23)) then corresponds the best, but its deviation is smaller compared to our model (Eq. (21)), which has the second-best match for smaller number of coefficients (3 parameters λ_0, λ_1, β). This proves the relevance of our model.

8. Discussion

The small variation of the capacitance at the beginning of the ageing process is certainly due to the insulating resin contained in the capacitor casing. The capacitors were not impacted by humidity at the beginning.

As illustrated in Fig. 9, the capacitance's degradation follows an exponential law with a failure rate of λ_0. Then, the acceleration phase of the degradation is a result of the diffusion. As a result, more and more areas of the metallization are affected by the corrosion. As the electrode surface is finite, the degradation process inevitably slows down. When the moisture has fully diffused, the capacitance degradation follows then an exponential law with a failure rate λ_1.

The model proposed in this paper fits well with the transition from λ_0 to λ_1. However, some discrepancies remain. We need to come back to our assumption to explain these discrepancies.

Firstly, it is very likely that the electrodes' thickness is not equal among all the electrodes. On one hand, their dimension on the order of nm, makes homogenization difficult during the manufacturing process. On the other hand, it is common for manufacturers to offer several electrode profiles, with strategically reinforced metallization. Some of these electrode profiles are sloped, so their thickness varies accordingly. The actual design of the electrode profile for each capacitor is therefore necessary to improve the accuracy of the model. However, this information is often difficult to obtain, either because of the required

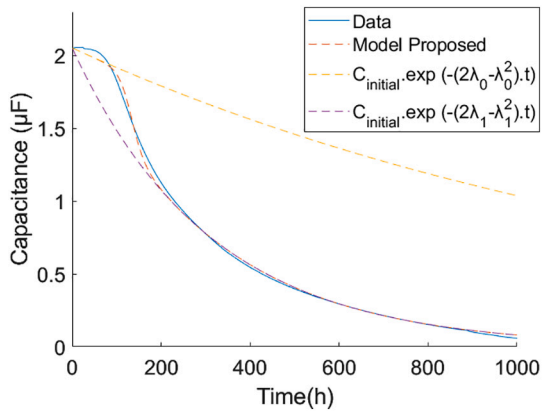


Fig. 9. Details of the fitting of the proposed model for one capacitor of the second test (cf. Table 1).

precision of the electrode profile measurement (<1 nm), or because of confidential manufacturing.

Secondly, the model proposed consider the diffusion homogenous on all the electrode's area, with a diffusion coefficient β constant. This assumption is not completely true. As seen on the films, the moisture intrudes preferentially from the edges next to the margins: the moisture diffusion is faster in this area. It could be explained by the air gap that exists at this point due to the absence of metallization. Moisture diffuses faster through the air than through the polypropylene. Impurities as air gaps at the interface between the polypropylene and the metallization enable similarly a faster access to the moisture to etch the metallization at some specific area.

Finally, the model considers that the failure rate of the surface unit depends solely on moisture diffusion. The impact of the failure of one surface unit on the other is not regarded. The corrosion could even gradually create "isolated metalized islands" that will be not connected anymore to the electrodes. The part of the area loss in the electrode is no longer a single surface unit but a whole set of surface units. However, even if this scenario is possible, it is very unlikely to happen. As the moisture tends to diffuse from the electrode margin to the schoopage, the first area unit to fail is the furthest one from the schoopage, but it does not disconnect the other ones from the power supply.

9. Conclusion

By considering metallized film capacitors, the existing laws in the literature are limited to their areas of focus. They model only a part of the phenomena and should be extended. The results of accelerated ageing tests under temperature, humidity and voltage identify the corrosion as the main failure mechanism. Therefore, based on the data collected, this paper proposes an alternative law to model the capacitance degradation caused by corrosion. Said model can be easily completed to include the self-healing phenomenon, in the same way by modelling the failure rate of the electrode surface unit, and therefore to be applicable to ageing. By predicting the ageing, the knowledge we have thanks to these ageing laws contributes to the maintenance requirements expectancy and the optimization of the capacitors use above 20 % of capacitance loss. However, this model does have its limitations. As mentioned above, it does not consider thickness or moisture heterogeneities along the electrode. Therefore, it does not predict the stabilization phase of the capacitance degradation which was observed during the ageing. One of the aims of this paper is to improve the model accordingly. It would also be relevant to assess its applicability for other ageing conditions.

CRediT authorship contribution statement

Claire Rochefort: Methodology, Formal analysis, Investigation, Writing – original draft, Visualization. **Pascal Venet:** Resources, Supervision, Writing – review & editing. **Guy Clerc:** Resources, Supervision, Writing – review & editing. **Ali Sari:** Resources, Supervision, Writing – review & editing. **Radoslava Mitova:** Resources, Supervision, Writing – review & editing. **Miao-Xin Wang:** Resources, Supervision, Writing – review & editing. **Pascal Bevilacqua:** Resources, Software. **Younes Zitouni:** Resources.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Rochefort Claire reports financial support was provided by Schneider Electric SE.

Data availability

The data that has been used is confidential.

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